

Representing Research Attention as Contextually Structured Flows

Jessica Rodrigues*, Angelo Salatino**, Gard Jensen***, and Scott Hale*

**jessica.rodrigues@oii.ox.ac.uk; scott.hale@oii.ox.ac.uk*
University of Oxford, United Kingdom

***angelo.salatino@open.ac.uk*
The Open University, United Kingdom

****gard.jensen@springernature.com*
Springer Nature, United Kingdom

Research attention is widely used as an indicator of visibility, influence, and societal uptake, yet it is typically represented as aggregated counts that do not preserve how attention develops across contexts over time. This creates a mismatch between how attention is interpreted and how it is represented. We propose attention flows as contextually structured representations that encode the organisation of attention and its evolution over time. We evaluate whether these representations capture transferable structure by constructing a benchmark based on analogy-style reasoning across research outputs. Comparing signal, sequence, and flow-based representations, we find that flow representations more effectively support structural comparison, particularly in settings where attention is shaped by temporal progression or context distributions. We further show that learned flow representations improve robustness under partial observation and structural perturbation. Overall, these results support modelling attention as a contextually structured phenomenon and provide a basis for more informative approaches to research evaluation.

1. Introduction

Attention, rather than information, has become the limiting resource shaping how ideas and artefacts circulate in contemporary communication systems (Simon, 1996; Davenport & Beck, 2001). Under conditions of informational abundance, attention concentrates unevenly, with only a small subset of objects attracting sustained collective engagement. This allocation becomes visible through communicative activity, as ideas and artefacts are referenced, discussed, and engaged with across contexts. In research, such activity takes the form of references, discussions, endorsements, critiques, and other modes of engagement, and is widely used to assess research contributions (Garfield, 1972; Bornmann, 2013; Sugimoto & Larivière, 2017; Hicks et al., 2015; Wouters et al., 2015). Crucially, this activity is not interpreted solely in terms of magnitude. Rather, it is understood through how attention accumulates, persists, and declines over time, reflecting broader patterns of diffusion and uptake (Rogers, 2003; Bornmann, 2012). In this way, attention informs judgements of visibility, relevance, influence, and uptake, which depend on the structure through which attention unfolds.

In practice, this activity is operationalised as attention signals, that is, contextually indexed records that make attention empirically observable. Because these signals are often treated as proxies for evaluative judgements, they shape how ideas and artefacts are perceived and acted upon in social contexts (Priem et al., 2011; Haustein et al., 2014). In research evaluation, this interpretation takes specific forms. Citations are treated as signals of intellectual influence, while mentions in policy documents or mainstream media indicate broader societal relevance. References in guidelines or institutional reports are taken as evidence of uptake, reflecting incorporation into practice. Within digital infrastructures, these signals constitute a widely used empirical unit through which attention is observed and compared across domains (Wouters et al., 2015; Sugimoto & Larivière, 2017; Wu & Huberman, 2007; Weng et al.,

2012). However, when attention is represented in this way, its underlying structure is not preserved.

This limitation becomes evident in concrete cases. During the COVID-19 pandemic, some research outputs rapidly attracted intense attention across media coverage, social media discussion, and policy debates, often in response to preliminary or contested findings (Gao et al., 2020; Mehra et al., 2020). In parallel, other lines of research accumulated attention more gradually over extended periods, reflecting sustained scientific development and eventual incorporation into applications such as vaccine design (Pardi et al., 2018). In both cases, attention became highly visible across media, policy, and public discourse, yet the ways in which it unfolded differed substantially. Some developments led to durable scientific and societal outcomes, while others were later questioned or retracted. When attention is represented solely through aggregated signals, these developments reduce to comparable counts of attention events. As a result, distinct developments may give rise to similar signal profiles, even when they reflect fundamentally different trajectories and outcomes. This reveals a representational mismatch: while attention is interpreted through how it unfolds over time, it is commonly represented as discrete observations that do not preserve this structure.

In response, we investigate whether attention can be represented as contextually structured objects. If attention exhibits latent structure, it should be reflected in how attention develops over time across communicative contexts. We refer to such objects as attention flows, defined as representations whose identity depends on the organisation of attention over time. This formulation follows principles from representation learning, which emphasise encoding structure that is constitutive of the phenomenon within representations (Bengio et al., 2013). We examine whether attention flows satisfy the conditions required to function as representational objects. Specifically, we assess whether structure is consistently preserved across representations, whether comparable developments exhibit similar structural patterns, and whether these patterns arise from the data itself rather than being imposed externally. Establishing these properties provides a basis for interpreting attention as a structured phenomenon, enabling meaningful comparison across developments and supporting more principled approaches to research evaluation.

2. Background

2.1 Attention signals as indicators

Bibliometric and alternative indicators are derived from observable traces of scholarly and societal activity, such as citations, policy mentions, and online engagement. Citations have long been used as proxies for scientific influence within research communities (Garfield, 1972), while more recent approaches incorporate broader forms of interaction with research outputs, including references in policy documents and mentions in digital environments (Bornmann, 2013; Sugimoto & Larivière, 2017). These developments have led to the emergence of alternative metrics that capture diverse forms of engagement beyond academic citation counts (Priem et al., 2011). In this context, online interactions such as tweets, blog posts, and news mentions are commonly interpreted as traces of attention to research outputs (Priem et al., 2011; Haustein et al., 2014). We refer to these observable traces as attention signals, which record instances in which ideas and artefacts attract attention across communicative contexts.

These indicators exhibit distinct temporal patterns. Studies of citation dynamics show that scientific influence evolves through phases of growth, peak visibility, and decline, often following regular and predictable trajectories (Wang, Song & Barabási, 2013). Similar patterns have been observed in online environments, where collective attention rises and decays over time, sometimes rapidly and in competition with other signals (Wu & Huberman, 2007; Weng et al., 2012). At the same time, different research developments may exhibit comparable temporal profiles, suggesting that attention follows recurrent structural patterns across domains and contexts. These observations indicate that research attention is not only a matter of magnitude, but also of temporal development, shaped by how attention accumulates, stabilises, and dissipates over time. Importantly, these patterns emerge from how attention is expressed across communicative contexts at different points in time, suggesting that attention may be characterised not only through individual signals, but through the evolving organisation of attention across contexts.

2.2 Representational limitations

Despite the temporal richness of attention signals, they are typically represented through structured data derived from observable events. In bibliometric and altmetric systems, citations, mentions, and interactions are operationalised as discrete observations that can be aggregated into counts or organised as time-indexed sequences (Priem et al., 2011; Haustein et al., 2014; Thelwall & Kousha, 2015). These representations enable large-scale analysis by providing standardised formats for measurement. In practice, attention is encoded either as collections of independent observations, such as total counts, or as ordered sequences reflecting when events occur over time (Moed, 2005). However, these forms do not explicitly encode relationships between observations beyond temporal ordering.

This limits the ability to capture how attention develops. While sequences preserve when events occur, they do not represent how attention is organised across communicative contexts at a given moment, nor how this organisation evolves. As a result, attention is not preserved as a coherent, structured development, and the relationships through which it accumulates, stabilises, or changes are not retained. Distinct developments may therefore appear similar when reduced to comparable counts or sequences, even when their underlying dynamics differ substantially. This concern aligns with broader critiques of research metrics, which highlight the loss of contextual and relational information in indicator-based evaluation (Wouters et al., 2015; Hicks et al., 2015; Bornmann, 2013). From a representation learning perspective, this suggests that the form of representation determines which regularities can be captured and analysed (Bengio et al., 2013), motivating the need for representations that preserve the structure through which attention unfolds.

3. Representing Attention as Flows

3.1 Distributional hypothesis

We approach attention as a phenomenon that requires representation in a form that preserves the structure through which it unfolds. In representation learning, the objective is to encode the factors of variation that are constitutive of a phenomenon, rather than incidental aspects of its observation (Bengio et al., 2013; Goodfellow et al., 2016). From this perspective, the choice of representation determines which regularities can be captured and compared.

To guide this choice, we adopt a distributional view of attention. In linguistics and natural language processing, the distributional hypothesis states that meaning is derived from the contexts in which a unit occurs (Harris, 1954; Firth, 1957), and underlies models that represent structure through patterns of co-occurrence (Mikolov et al., 2013; Pennington et al., 2014). By analogy, we assume that the structure of attention can be inferred from how it is distributed across communicative contexts over time. This view is consistent with empirical studies showing that collective attention evolves through structured temporal processes across competing contexts (Wu & Huberman, 2007; Weng et al., 2012; Wang et al., 2013; Hausteine et al., 2014; Sugimoto et al., 2017).

Under this assumption, attention is not defined by individual signals, but by context distributions and their temporal evolution. Developments that exhibit similar distributions and trajectories correspond to comparable forms of attention, while differences reflect distinct dynamics. Standard representations, however, encode attention as independent observations or ordered sequences, without preserving how it is jointly distributed across contexts at a given time. This motivates representing attention as contextually structured distributions, forming the basis for the attention flow representation introduced next.

3.2 Attention Flow Representation

Based on this perspective, we define attention flows as contextually structured representations of attention. An attention flow is a sequence of attention states, where each state corresponds to a context distribution at a given point in time. Operationally, each state is represented as a context vector, where each dimension corresponds to a communicative context and values indicate the volume of attention observed in that context. An attention flow is then a sequence of such vectors indexed by discrete temporal intervals, capturing how attention is distributed across contexts and how this distribution evolves. This representation defines a space in which each state corresponds to a context distribution, and each flow corresponds to a sequence of such states.

However, while this construction preserves structure, it does not by itself ensure that relations between flows can be compared in a consistent way. To enable such comparisons, flows are mapped into a shared embedding space using predictive sequence models, where context distributions are encoded into latent states and transitions are learned through a mechanism that captures temporal dependencies. We consider several training objectives, including forward, bidirectional, and contrastive variants, learned using mini-batch optimisation in a fixed-dimensional latent space. In this space, states are encoded in a unified representation, allowing relations between attention developments to be expressed through vector-based operations. This enables comparisons across cases in terms of transformations rather than isolated observations, supporting reasoning over attention dynamics (Mikolov et al., 2013; Bengio et al., 2013; Battaglia et al., 2018).

3.3 Structural properties of attention flows

Having established a representation in which attention developments can be compared through vector-based operations, we specify the conditions under which such comparisons are meaningful. In representation learning, these conditions are typically framed in terms of invariance and the ability to encode structure rather than artefacts of observation (Bengio et al., 2013; Goodfellow et al., 2016). We define three properties: structural preservation, which requires that the representation maintains the relationships between attention states within a

case; structural invariance, which requires that comparable developments exhibit similar patterns across cases; and structural intrinsicness, which requires that the observed structure arises from the data rather than from the representation itself.

We assess these properties using vector operations that capture transformations between attention developments:

- Linear translation: $D \approx A - B + C$
- Relational similarity: $(A - B) \approx (D - C)$
- Counterfactual transfer: $D \approx C + (B - A)$

These operations follow prior work showing that meaningful representations encode structure through patterns that support comparison and reasoning (Mikolov et al., 2013; Gentner, 1983; Bengio et al., 2013; Battaglia et al., 2018).

4. Empirical Evidence of Attention Flows

4.1 Data

We construct a dataset of contextually indexed attention signals associated with research outputs, operationalising the concepts introduced in Sections 2 and 3 as concrete data objects. The analysis uses data from the Scientometric Researcher Access to Data (SRAD) programme, which provides access to the Altmetric and Dimensions databases. Altmetric aggregates publicly observable traces of interaction with research outputs across online environments, including social media, news, blogs, and policy (Priem et al., 2011; Haustein et al., 2014), while Dimensions provides bibliographic metadata and persistent identifiers that enable signals to be consistently linked to research outputs (Moed, 2005). We restrict the analysis to research articles published between 1 January 2016 and 1 January 2024 in the Computer Science domain (FoR codes 4601–4613).

Each attention signal is defined as a tuple (DOI, context, date, count), capturing observed interactions in a given context and day. Signals are indexed at daily resolution and restricted to post-publication activity, forming time series of attention across contexts such as social media, news, and policy (Table 1), comprising 472,917 signals across 43,520 outputs. To construct representation-ready data, we apply a three-step pipeline: **(1) case selection**, retaining outputs with sufficient temporal extent, contextual diversity, and dynamic variation, yielding 178,073 signals across 7,355 cases; **(2) temporal structuring** into ordered daily sequences; and **(3) contextual aggregation**, where each time step corresponds to a distribution over contexts. From these sequences, we derive two baseline representations: a signal representation that reduces each case to total attention volume, removing temporal and contextual structure, and a sequence representation that models attention as an ordered series of daily counts, preserving temporal dynamics while collapsing contextual variation.

Table 1. Example attention signal records for a single research output.

DOI	Date	Context	Count
10.1038/...	2016-01-28	news	48
10.1038/...	2016-01-28	blogs	7
10.1038/...	2016-01-28	facebook	6
10.1038/...	2016-01-29	twitter	12
10.1038/...	2016-01-30	reddit	5
10.1038/...	2016-02-01	policy	2

4.2 Benchmark construction

To evaluate whether attention representations encode transferable structure, we construct a structural reasoning benchmark from the held-out test partition. The benchmark consists of analogy-style queries of the form $A:B::C:D$, testing whether transformations between cases can be recovered from a representation space. This formulation follows prior work showing that meaningful embeddings support vector-based reasoning and encode structure through consistent transformations (Mikolov et al., 2013; Bengio et al., 2013; Battaglia et al., 2018). To operationalise this, we define an external reference embedding space based on interpretable case-level features capturing temporal dynamics, context distribution, and scale, providing a structured and independent description of attention dynamics.

We evaluate representations under controlled structural conditions by defining five reasoning regimes: relational, sequential, distributional, diffuse, and adversarial. The relational regime captures generic transformations without a dominant structural pattern. The sequential regime focuses on transformations that depend on temporal progression, requiring the preservation of ordered dependencies. The distributional regime captures changes in context distributions across shared environments. In contrast, the diffuse regime represents weak or under-specified structure, while the adversarial regime introduces complex dynamics, providing a stress test for robustness.

Within each regime, we construct queries using the reference embedding space. For each pair (A, B) , we sample cases C and identify a target D such that the transformation from A to B is matched by that from C to D , following vector-offset reasoning. Candidate queries are filtered using thresholds on reconstruction error and relative fit to ensure well-defined transformations. This yields 110,462 valid queries, from which we construct a balanced benchmark by sampling up to 5,000 queries per regime, resulting in 20,031 queries stratified by difficulty. This benchmark provides a controlled setting to assess whether different representations recover transformations across varying structural conditions.

4.4 Experiments

4.4.1 Structural invariance

We begin by asking whether representations encode structure that generalises across attention developments. If this property holds, transformations observed between one pair of cases should transfer to others, enabling comparison through shared patterns rather than individual observations. To test this, we use the retrieval benchmark introduced in Section 4.3, which organises cases into five structural regimes. For each triplet (A, B, C) , we construct a target representation using the vector operation $A - B + C$ and retrieve the most similar candidate D using cosine similarity. Performance is measured using mean relative rank (MRR), computed separately for each regime.

Tables 2 and 3 report retrieval performance for non-parametric and learned representations. Signal and sequence representations achieve comparable overall performance (0.49 vs 0.50), indicating that encoding temporal order alone provides limited gains. In contrast, sparse flow representations improve performance across all regimes, reaching an overall MRR of 0.52, with the largest gains in the adversarial (0.55) and sequential (0.51) regimes. Learning further enhances these results: all objectives outperform the sparse baseline, increasing overall MRR to 0.53 (forward), 0.55 (bidirectional), and 0.56 (contrastive). The contrastive objective

performs best across regimes, with consistent improvements in both well-structured settings (e.g. distributional: 0.54) and more challenging ones (e.g. diffuse: 0.54, adversarial: 0.53), indicating that a shared embedding space makes structural patterns more accessible for comparison.

Table 2. Retrieval performance (MRR) of non-parametric representations across structural regimes.

Representation	Relational	Sequential	Distributional	Diffuse	Adversarial	Overall
Signal	0.501	0.534	0.491	0.514	0.418	0.492
Sequence	0.526	0.507	0.548	0.549	0.438	0.501
Flow (sparse)	0.530	0.511	0.518	0.526	0.555	0.523

Table 3. Retrieval performance (MRR) of learned flow representations across structural regimes.

Representation	Relational	Sequential	Distributional	Diffuse	Adversarial	Overall
Flow (Forward)	0.523	0.482	0.474	0.517	0.499	0.532
Flow (Bidirectional)	0.524	0.470	0.492	0.522	0.489	0.551
Flow (Contrastive)	0.581	0.512	0.647	0.641	0.532	0.564

Figure 1 (**distributional regime**) shows stable co-occurrence patterns across shared contexts (e.g. X, Reddit, News, Blogs, Patent), where all cases occupy a common communicative space but differ in how attention is distributed within it. The relation between A and B corresponds to a shift in emphasis across these contexts, rather than a change in their ordering or presence. Applied to C, the retrieved case D reflects a comparable redistribution, aligning with the same contextual scaffold while reweighting attention across platforms. This indicates that the analogy operates over relative prominence within a shared environment, a pattern that is not visible in aggregated representations and only weakly captured by sequences.

Figure 2 (**sequential regime**) highlights transformations in temporal ordering. All cases follow a progression from early visibility to later consolidation across contexts such as X, Wikipedia, Blogs, Policy, and News, but differ in how this trajectory is organised over time. The relation between A and B reflects a reordering of this progression, shifting the timing and prominence of intermediate stages. Applied to C, the retrieved case D preserves this ordered development, maintaining a similar transition from initial exposure to later-stage consolidation. Here, the analogy depends not on where attention accumulates, but on how it unfolds, showing that temporal structure is essential for capturing comparable developments.

Figure 3 (**adversarial regime**) presents a more complex case in which all developments exhibit highly similar and richly structured trajectories. A corresponds to the AlphaFold paper and B to a paper that explores how large language models encode clinical knowledge. The relation between them is not tied to a single context or temporal shift, but reflects a broader transformation from foundational scientific discovery to applied, system-level knowledge integration. Applied to C, which reports a high-impact neuroscience finding on bird forebrain neuron density, the retrieved case D preserves a similar pattern of broad diffusion and downstream uptake. This suggests that, even in adversarial settings, flow-based representations can capture semantic regularities that operate at a more abstract level than surface structure.

Figure 1. Analogy-based retrieval in the distributional regime: $A \rightarrow B$ applied to C yields D.

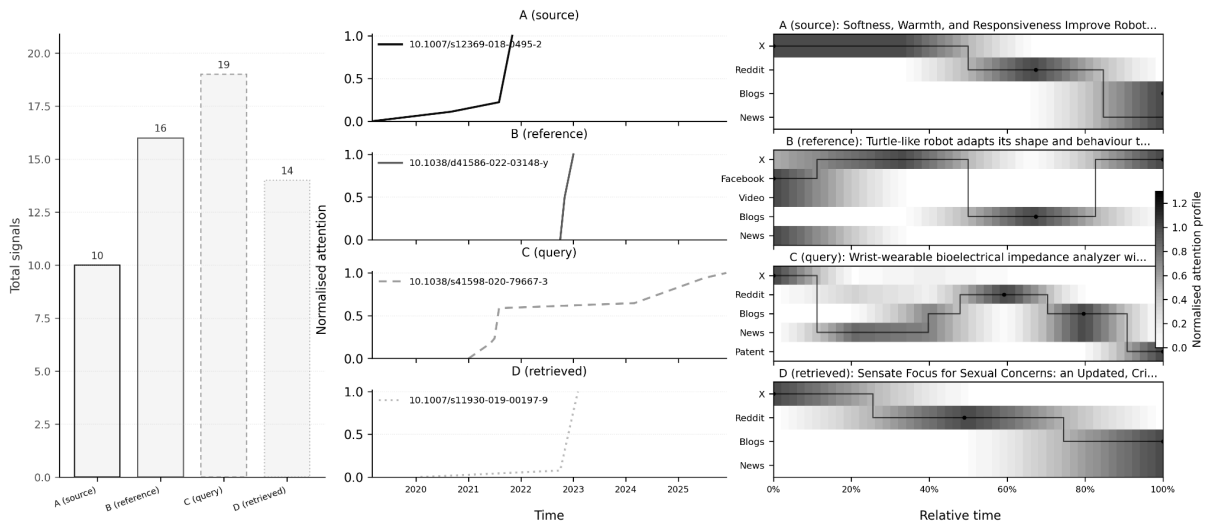


Figure 2. Analogy-based retrieval in the sequential regime: $A \rightarrow B$ applied to C yields D.

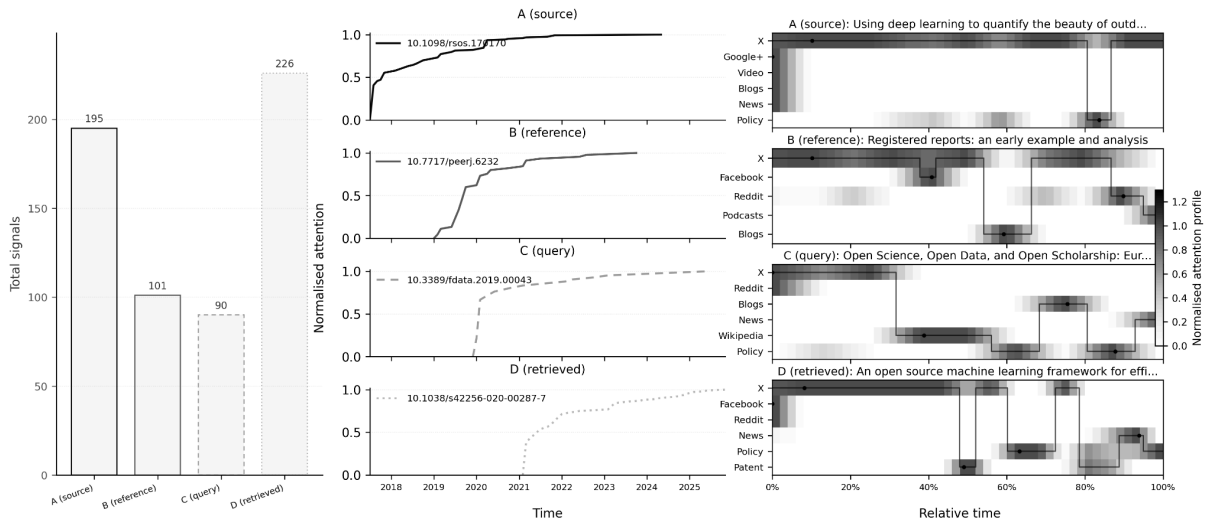
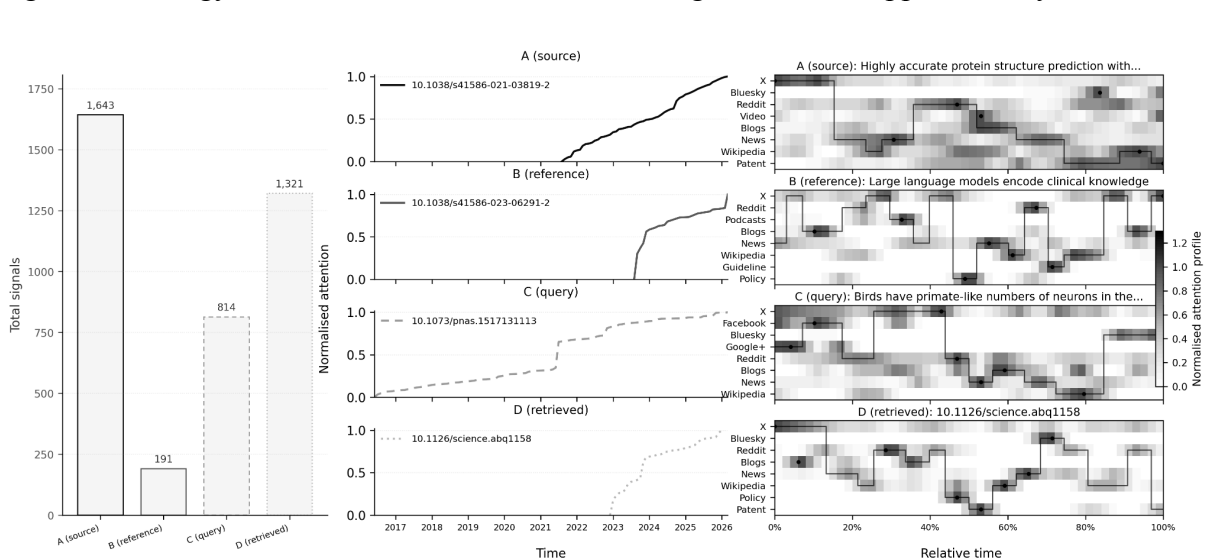


Figure 3. Analogy-based retrieval in the adversarial regime: $A \rightarrow B$ applied to C yields D.



4.4.2 Structural preservation

Structural preservation evaluates whether representations retain meaningful structure under partial observation, allowing transformations inferred from early stages to remain consistent with complete trajectories. We assess this by truncating the query case C to 25%, 50%, and 75% of its development while keeping candidate cases fully observed. Retrieval performance is measured using mean relative rank (MRR) across regimes.

Table 4 reports results across all representations. Performance declines as less of the trajectory is observed, with the sharpest drop at 25% truncation. Signal and sequence representations degrade substantially across all regimes, indicating limited robustness to partial observation. In contrast, flow-based representations degrade more gradually, maintaining higher performance at each truncation level. This advantage is most pronounced in the distributional and sequential regimes, where early segments already reflect how attention is organised across contexts and time. Learned flow variants further improve robustness over the sparse baseline, with the contrastive objective achieving the strongest overall performance and exhibiting limited degradation even under severe truncation. These results indicate that flow representations encode structural patterns that remain identifiable from incomplete trajectories.

Table 4. Retrieval performance (MRR) under truncation across regimes.

Representation	Regime	Truncation (25%)	Truncation (50%)	Truncation (75%)
Signal	Distributional	0.492	0.492	0.492
	Sequential	0.492	0.492	0.492
	Diffuse	0.492	0.492	0.492
	Adversarial	0.492	0.492	0.492
Sequence	Distributional	0.549	0.574	0.565
	Sequential	0.503	0.490	0.482
	Diffuse	0.545	0.561	0.563
	Adversarial	0.549	0.565	0.568
Flow (sparse)	Distributional	0.520	0.535	0.535
	Sequential	0.510	0.507	0.511
	Diffuse	0.513	0.524	0.527
	Adversarial	0.517	0.528	0.529
Flow (Forward)	Distributional	0.499	0.513	0.515
	Sequential	0.516	0.490	0.482
	Diffuse	0.498	0.506	0.508
	Adversarial	0.521	0.523	0.524
Flow (Bidirectional)	Distributional	0.503	0.515	0.513
	Sequential	0.516	0.494	0.482
	Diffuse	0.501	0.509	0.508
	Adversarial	0.503	0.505	0.505
Flow (Contrastive)	Distributional	0.579	0.576	0.593
	Sequential	0.491	0.489	0.495
	Diffuse	0.592	0.582	0.599
	Adversarial	0.595	0.585	0.601

4.4.3 Structural intrinsicness

Structural intrinsicness evaluates whether the structure captured by a representation arises from the data rather than from artefacts of the representation. We assess this by perturbing the query cases (A, B, C), randomly shuffling the internal ordering of attention signals while

preserving overall magnitude. Retrieval is then performed using the same setup, and performance is measured using mean relative rank (MRR) across regimes.

Table 5 reports results across all representations. Signal representations are largely unaffected by perturbation, reflecting their reliance on aggregate counts rather than internal structure. In contrast, sequence representations exhibit clear performance degradation, indicating sensitivity to temporal ordering. Flow-based representations also degrade consistently, with the strongest effects in the sequential regime, where structure depends on ordered transitions, and smaller but still noticeable drops in the distributional regime. Learned flow variants maintain higher performance than non-parametric baselines, suggesting that they capture more stable structural regularities, though they remain sensitive to perturbation. These results confirm that flow representations encode intrinsic structure that depends on the organisation of attention rather than static properties.

Table 5. Retrieval performance (MRR) under perturbation across regimes.

Representation	Regime	MRR (Original)	MRR (Perturbed)	Δ MRR ↓
Signal	Distributional	0.492	0.492	0.000
	Sequential	0.492	0.492	0.000
	Diffuse	0.492	0.492	0.000
	Adversarial	0.492	0.492	0.000
Sequence	Distributional	0.551	0.530	-0.021
	Sequential	0.473	0.473	-0.001
	Diffuse	0.536	0.520	-0.016
	Adversarial	0.531	0.515	-0.011
Flow (sparse)	Distributional	0.535	0.516	-0.019
	Sequential	0.507	0.499	-0.008
	Diffuse	0.527	0.512	-0.015
	Adversarial	0.520	0.451	-0.008
Flow (Forward)	Distributional	0.520	0.502	-0.018
	Sequential	0.475	0.472	-0.002
	Diffuse	0.503	0.495	-0.008
	Adversarial	0.502	0.494	-0.007
Flow (Bidirectional)	Distributional	0.516	0.501	-0.016
	Sequential	0.476	0.473	-0.004
	Diffuse	0.505	0.493	-0.012
	Adversarial	0.503	0.491	-0.010
Flow (Contrastive)	Distributional	0.594	0.531	-0.063
	Sequential	0.494	0.493	-0.002
	Diffuse	0.603	0.545	-0.058
	Adversarial	0.601	0.543	-0.056

5. Discussion

5.1 Implications for indicators

Our findings point to a limitation of signal-based indicators. While attention is interpreted through its development over time, standard representations reduce it to aggregated or sequential observations that do not preserve its structure. As a consequence, they capture magnitude and, to some extent, temporal variation, but do not encode how attention is organised across contexts or how this organisation evolves. Flow-based representations address this limitation by retaining the structure of attention as it unfolds, allowing developments to be analysed as coherent processes rather than collections of events.

This shift has direct implications for research evaluation. When attention is reduced to counts or sequences, developments that differ in how they accumulate, stabilise, and propagate may appear comparable. By contrast, representing attention as flows makes it possible to distinguish between transient visibility and sustained developments that extend across contexts and over time. More broadly, this supports a move from measuring attention as accumulation to modelling it as a process, where evaluation depends on the organisation and progression of attention rather than its volume alone. In this sense, attention flows provide a basis for indicators that are aligned with how attention is interpreted in practice.

5.2 Limitations and future work

At the same time, the approach is subject to several limitations. Attention signals are derived from observable digital traces and therefore provide a partial and potentially biased view of engagement across contexts and audiences. Moreover, the construction of attention flows depends on design choices in temporal aggregation, context definition, and representation learning, each of which shapes the structure that is captured. The evaluation further relies on analogy-based retrieval as a proxy for structural comparison, which, while informative, does not directly measure downstream impact.

Addressing these limitations requires extending both the data and the modelling framework. Incorporating additional sources of attention, including expert assessment and institutional signals, would provide a more complete view of engagement. At the same time, richer representations that integrate textual, relational, or citation-based information may improve the modelling of attention dynamics. Finally, establishing clearer links between attention flows and downstream outcomes would enable more direct evaluation of societal impact. Taken together, these directions point toward a broader programme in which attention is analysed not only as observable activity, but as a structured process connecting research to its wider effects.

Open science practices

This study uses data obtained through the Scientometric Researcher Access to Data (SRAD) programme, including Altmetric and Dimensions. While these sources provide large-scale and high-quality coverage of attention signals, they are not fully open, and access is subject to licensing restrictions. For this reason, the raw data cannot be publicly shared. To support transparency, we provide a detailed account of the data collection, filtering, and preprocessing pipeline.

All derived artefacts are made openly available. This includes the structural reasoning benchmark introduced in this work, as well as the source code used to construct representations, train models, and run evaluations, available at <https://github.com/jehrodrigues/representing-research-attention-as-contextually-structured-flows>.

These resources enable full reproducibility of the experimental setup and support reuse and extension. No formal preregistration was conducted, as the study is exploratory, but all methodological choices are explicitly specified.

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Author contributions

The first author led the conceptualisation of the study, including the formulation of the attention flow representation and the overall research design. She was responsible for data curation, methodology development, implementation of models and experiments, formal analysis, and writing of the manuscript. The remaining authors provided supervision and advisory input on theoretical framing, modelling choices, evaluation design, and the scientometric context, and contributed through critical feedback on the manuscript.

Competing interests

The authors declare that they have no competing interests.

References

Barabási, A.-L. (2016). Network science. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9781107415324>

Battaglia, P.W., Hamrick, J.B., Bapst, V., Sanchez-Gonzalez, A., Zambaldi, V., Malinowski, M., et al. (2018). Relational inductive biases, deep learning, and graph networks. arXiv preprint arXiv:1806.01261. <https://arxiv.org/abs/1806.01261>

Belinkov, Y. (2022). Probing classifiers: Promises, shortcomings, and advances. Computational Linguistics, 48(1), 207–219. https://doi.org/10.1162/coli_a_00422

Belinkov, Y., Bisk, Y. (2018). Synthetic and natural noise both break neural machine translation. In Proceedings of the International Conference on Learning Representations (ICLR)

Bengio, Y., Courville, A., Vincent, P. (2013). Representation learning: A review and new perspectives. IEEE Transactions on Pattern Analysis and Machine Intelligence, 35(8), 1798–1828. <https://doi.org/10.1109/TPAMI.2013.50>

Bornmann, L. (2013). What is societal impact of research and how can it be assessed? A literature survey. Journal of the American Society for Information Science and Technology, 64(2), 217–233. <https://doi.org/10.1002/asi.22803>

Ethayarajh, K. (2019). How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 embeddings. In Proceedings of EMNLP, 55–65

Firth, J.R. (1957). A synopsis of linguistic theory 1930–1955. In Studies in linguistic analysis (pp. 1–32). Oxford: Blackwell

Garfield, E. (1972). Citation analysis as a tool in journal evaluation. Science, 178(4060), 471–479. <https://doi.org/10.1126/science.178.4060.471>

Gentner, D. (1983). Structure-mapping: A theoretical framework for analogy. Cognitive Science, 7(2), 155–170. https://doi.org/10.1207/s15516709cog0702_3

Goodfellow, I.J., Shlens, J., Szegedy, C. (2015). Explaining and harnessing adversarial examples. In Proceedings of the International Conference on Learning Representations (ICLR)

Goodfellow, I., Bengio, Y., Courville, A. (2016). Deep learning. Cambridge, MA: MIT Press

Harris, Z.S. (1954). Distributional structure. *Word*, 10(2–3), 146–162. <https://doi.org/10.1080/00437956.1954.11659520>

Haustein, S., Peters, I., Sugimoto, C.R., Thelwall, M., Larivière, V. (2014). Tweets as impact indicators: Examining the implications of automated “bot” accounts on Twitter. *Journal of the Association for Information Science and Technology*, 65(2), 232–238. <https://doi.org/10.1002/asi.23103>

Hewitt, J., Manning, C.D. (2019). A structural probe for finding syntax in word representations. In *Proceedings of NAACL-HLT*, 4129–4138. <https://doi.org/10.18653/v1/N19-1419>

Hicks, D., Wouters, P., Waltman, L., de Rijcke, S., Rafols, I. (2015). The Leiden Manifesto for research metrics. *Nature*, 520(7548), 429–431. <https://doi.org/10.1038/520429a>

Lazer, D., Pentland, A., Adamic, L., Aral, S., Barabási, A.-L., Brewer, D., et al. (2009). Computational social science. *Science*, 323(5915), 721–723. <https://doi.org/10.1126/science.1167742>

Mikolov, T., Yih, W.-t., Zweig, G. (2013). Linguistic regularities in continuous space word representations. In *Proceedings of NAACL-HLT*, 746–751

Moed, H.F. (2005). *Citation analysis in research evaluation*. Dordrecht: Springer. <https://doi.org/10.1007/1-4020-3714-7>

Pennington, J., Socher, R., Manning, C.D. (2014). GloVe: Global vectors for word representation. In *Proceedings of EMNLP*, 1532–1543. <https://doi.org/10.3115/v1/D14-1162>

Priem, J., Taraborelli, D., Groth, P., Neylon, C. (2011). *Altmetrics: A manifesto*. <http://altmetrics.org/manifesto/>

Sugimoto, C.R., Larivière, V. (2017). *Measuring research: What everyone needs to know*. Oxford: Oxford University Press. <https://doi.org/10.1093/oso/9780190640120.001.0001>

Sutskever, I., Vinyals, O., Le, Q.V. (2014). Sequence to sequence learning with neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*

Thelwall, M., Kousha, K. (2015). Web indicators for research evaluation. *Journal of Informetrics*, 9(4), 785–794. <https://doi.org/10.1016/j.joi.2015.07.007>

Wang, D., Song, C., Barabási, A.-L. (2013). Quantifying long-term scientific impact. *Science*, 342(6154), 127–132. <https://doi.org/10.1126/science.1237825>

Weng, L., Flammini, A., Vespignani, A., Menczer, F. (2012). Competition among memes in a world with limited attention. *Scientific Reports*, 2, 335. <https://doi.org/10.1038/srep00335>

Wouters, P., et al. (2015). The metric tide: Literature review. HEFCE.
<https://doi.org/10.13140/RG.2.1.5066.3520>

Wu, F., Huberman, B.A. (2007). Novelty and collective attention. Proceedings of the National Academy of Sciences, 104(45), 17599–17601. <https://doi.org/10.1073/pnas.0704916104>